

AGILE SOFTWARE ENGINEERING (CS303)

Complete Syllabus

Course Objectives

Understand Agile methodologies, Scrum practices, Agile requirements, XP practices and Agile metrics.

Unit 1 - Introduction to Agile

Agile Manifesto, Agile Principles, Waterfall vs Agile.

Unit 2 - Scrum Framework

Scrum Roles, Scrum Events, Sprint Planning, Product Backlog, Sprint Backlog.

Unit 3 - Agile Requirements

User Stories, Story Points, Agile Requirements, Acceptance Criteria.

Unit 4 - Extreme Programming

Pair Programming, TDD, Refactoring.

Unit 5 - Agile Metrics & CI/CD

Burn-down Charts, Agile Metrics, Continuous Integration, Continuous Delivery, CI/CD Pipelines.

Course Outcomes

Apply Agile practices in software projects and evaluate Agile development approaches.